

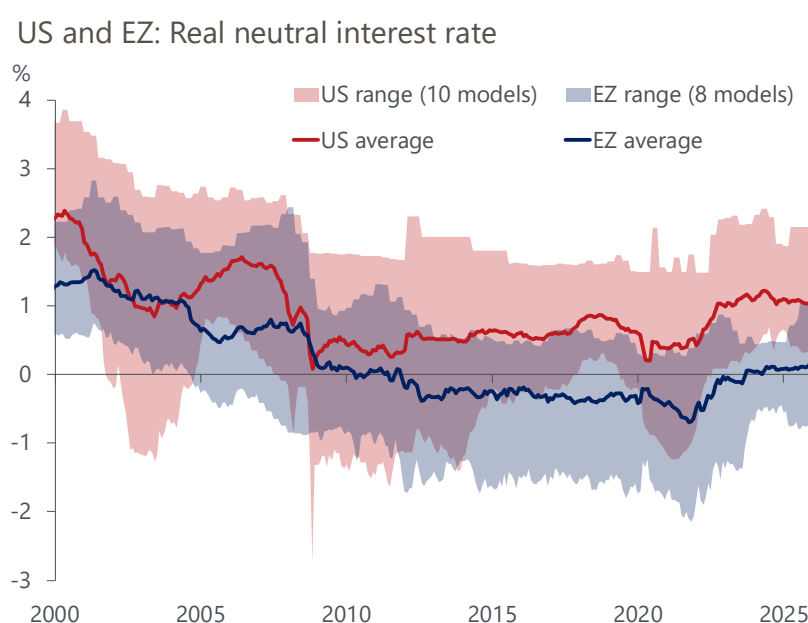
Research Briefing | Global

Neutral interest rates differ for a good reason

- The US Fed funds rate is significantly higher than the ECB's deposit rate, but that doesn't mean the Federal Reserve will cut rates to the same level in the long-run. The reason is that the neutral interest rate, or r^* – the equilibrium level at which the policy rate should eventually settle – is significantly lower in the Eurozone than it is in the US.
- In real terms, r^* is about 1% in the US and 0% in the Eurozone, according to an average of various macro- and financial market-based models. This aligns with our baseline forecast; however, our dynamic latent factor model indicates potential downside risks of up to 0.4ppts.
- While the exact level of r^* is highly uncertain, the US-Eurozone gap of 1ppt emerges as a robust finding in our analysis. Our structural general equilibrium model shows that the gap is the result of three uncontroversial factors: worse demographics and productivity growth in the Eurozone and a sharper rise in US government debt since 2000.
- Demographic change will remain a drag on r^* across major advanced economies, but improvements in productivity, continued AI investment, and quantitative tightening will nudge rates higher. According to our structural model, the net impact will range from around 0.3ppts in the Eurozone to 0.8ppts in the US by 2030.

As Fed Chair Jerome Powell's term runs out, a key question is how much further the Fed will lower policy rates in the future. Comparisons to the much lower rates in the Eurozone overlook cyclical and structural differences between the two economies. Our analysis shows that the structural differences mean that the long-term equilibrium rate r^* is much lower in the Eurozone (**Chart 1**), so we expect policy rates in the US to settle at a higher level.

Chart 1: The neutral interest rate is significantly lower in the Eurozone than in the US



Sources: Oxford Economics, Haver Analytics, ECB, others (**Box 1**)

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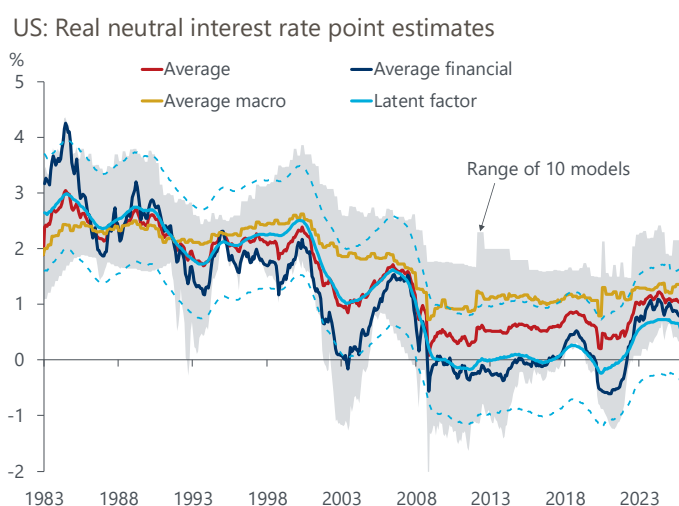
Uncertainty around the level of r^* , but a robust US-EZ gap

The level of the real neutral rate is highly uncertain, as it's not observable and must be inferred from data using models. For the US, the 10 models we track span values from 0.3% to 2.1% at end-2025, with an average of 1%. For the Eurozone, the eight models we track span values from -0.7% to 1.2%, with an average of 0.1%. These model averages align well with our baseline forecast of where policy rates in the two economies will settle in the medium- to long-run.

The models cover a wide range of data and approaches, so taking an average is like acknowledging that each has its merits (**Box 1**). Models based on macro data tend to focus more on longer-run trends, and are on average higher (**Chart 2 and 3**). Financial markets-based models tend to be more affected by shorter-run developments, are more volatile, and are typically lower on average. We also include surveys of forecasters and market participants in our range of models.

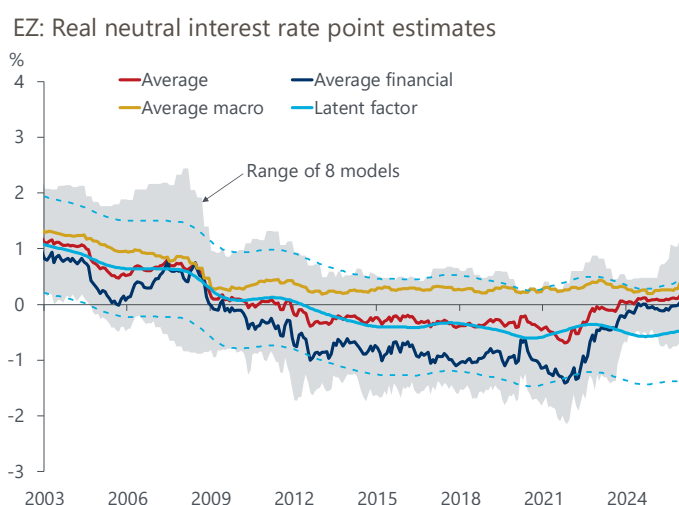
The gap between average financial and macro estimates might be interpreted as the extent to which r^* is expected to rise in the future. While this is by no means a scientific interpretation, it's in line with the results of our forward-looking structural model that we'll turn to below.

Chart 2: Our latent factor estimate is lower than the average of 10 model estimates



Sources: Oxford Economics, Haver Analytics, others (**Box 1**)

Chart 3: ... also for Eurozone r^* , where we have a shorter time series



Sources: Oxford Economics, Haver Analytics, others (**Box 1**)

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However, these simple averages also have some drawbacks. They can be distorted by outliers, currently the well-known [Lubik-Matthes model](#) in the US, at 2.15%, and the new [Christensen-Mouabbi model](#) in the Eurozone, at 1.17%. While that could be fixed with a median, we instead estimate a dynamic latent factor model, which can handle mixed frequencies and the addition of series with later starting points. Moreover, it distinguishes between signal and noise in each series at every point in time, 'learning' the informational content of each series over time. The latent factor can be interpreted as the common, underlying r^* estimate driving the various individual series.

Our latent factor model suggests downside risks to our baseline terminal policy rate forecast. It points to a US r^* of 0.6% and a Eurozone r^* of -0.4%, roughly 0.4ppts lower than the respective model averages. That's because it puts less weight on individual movements that are untypical of a series, and more weight on common movement and long-standing series. However, the latent factor estimates are currently at the low end of the range of models, an untypical situation that we think will reverse to some extent. An additional caveat is that the time series in the Eurozone may still be a bit short for the model to learn the properties of each individual series.

Perhaps our most striking finding is how robust the current gap between the US and Eurozone neutral rate is, at around 1ppt. This is independent of whether we use the total, financial, or the macro averages; the max and min of the range; or the latent factor. That means that lower policy rates in Europe are to be expected, as they're caused by macroeconomic forces. Using Europe's policy rates to argue for lower US rates is fundamentally flawed.

The US-EZ r^* gap is not new either, having persistently widened since the global financial crisis. We explain the forces driving this gap using [our established structural general equilibrium model](#).

The secular forces driving the neutral rates and the US-EZ gap

Since 2000, the main secular force pushing neutral rates down was demographic change, consisting of rising longevity, falling fertility rates, and changing migration patterns (**Chart 4**). To understand why, let's consider a capital markets framework where r^* equilibrates the aggregate supply of and aggregate demand for capital. Higher life expectancy makes people save more for retirement, all else equal, raising savings. Lower fertility rates mean fewer workers in the future, lowering expected GDP growth and dampening firms' investment. All these channels push down the neutral rate.

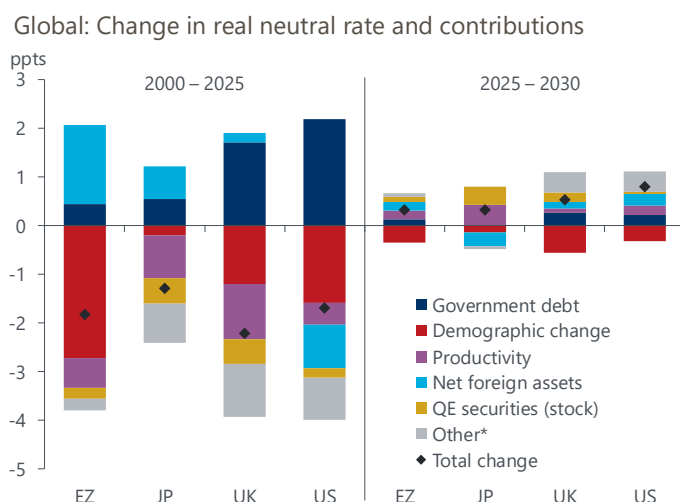
Japan has undergone the demographic transition much earlier than other major advanced economies, and consequently, the downward pressure on r^* since 2000 has been much more muted.

The other key factor that consistently depressed neutral rates was the slowdown in productivity growth, which affects r^* by lowering GDP growth and the expected return on investment.

These downward forces have been partially offset by rising government debt, which raised the demand for capital, or loanable funds more broadly, particularly in the US and the UK. In the Eurozone and Japan, capital outflows – shown as an increase in net foreign assets in **Chart 4** – have also mitigated the drop in r^* .

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Chart 4: Past falls in r^* will only be partially reversed over the next five years



Source: Oxford Economics

*rel. price of investment goods, labour share, consumer credit, GE interactions

Over the next five years, we forecast a slight increase in real neutral rates across advanced economies. While the reversal is quite modest – ranging from 0.3ppts for the Eurozone to 0.8ppts for the US – it does mark a notable break with the downward trend over recent decades.

While demographic change will remain a drag, improvements in productivity growth, rising private investment, and the reduction of securities held directly by central banks will all tilt the capital markets equilibrium in favour of higher rates. The new EU and German infrastructure and military spending will have only a marginal upward impact as EU government debt is limited by the bloc's fiscal rules. In the US, the favourable fertility trajectory is offset by a significant reduction in immigration. Lastly, in our baseline we forecast that AI-related investment will lead to a rise in the price of investment goods relative to consumption goods, particularly in the US (subsumed in 'other' in [Chart 4](#)). This also puts upward pressure on r^* because the supply of capital diminishes in real terms.

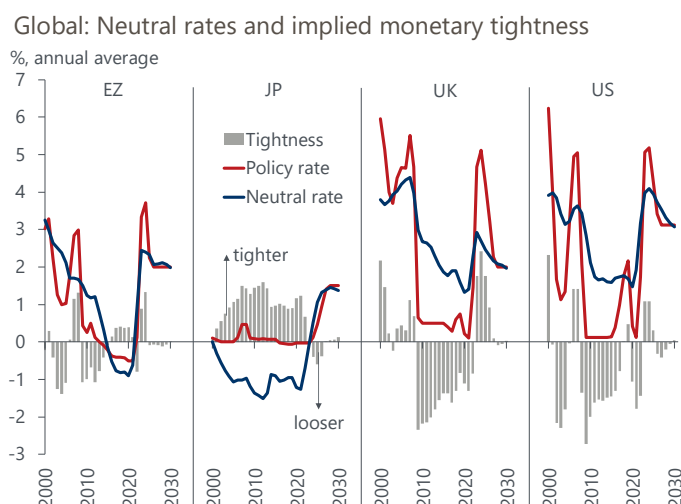
It's worth noting that our predictions for the Eurozone and US align quantitatively with the difference in longer-term macro estimates and shorter-term market-based estimates that we've highlighted above.

The implied monetary policy stance differs by economy

With our structural general equilibrium model, we can forecast the level of the neutral rate that's consistent with our monthly baseline forecast of the main structural forces. [Chart 5](#) shows this estimated neutral rate in nominal terms out to 2030 for four major economies, plotted against our policy rate forecast. The difference is an indication of monetary policy tightness, i.e., the more the policy rate is above the neutral rate, the tighter is the monetary policy stance, and vice versa.

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Chart 5: Policy rates will converge to nominal neutral rates at different speeds



Sources: Oxford Economics, Haver Analytics

According to our estimates, the ECB's main policy rate is [already close to neutral](#). In the US, the current monetary stance is [still moderately tight](#), in line with the objective of bringing inflation down without endangering employment, but we project the Fed funds rate to converge to neutral by end-2026. We'll discuss the monetary stance of the Bank of Japan separately in an upcoming briefing.

Box 1: Summary of neutral interest rate models for the eurozone (Chart 1)

Here we summarise the range of models shown in [Chart 1](#) and [Chart 3](#). We discussed the US models and our dynamic latent factor model in an [earlier US briefing](#) and focus here on the Eurozone models. Where models are estimated in nominal terms, we subtract model-consistent inflation expectations as explained below.

Macro models

- [Holston-Laubach-Williams \(2017\)](#): semi-structural model using the Phillips curve and investment-savings equation. Data: GDP, inflation, nominal interest rate. Decomposition into potential growth and 'other factors'.
- [Ferreira-Lott-Richards \(2025\)](#): state-space model using trend productivity, demographics, global supply of safe assets, demand factors for safe assets, and global spillovers.
- [Krippner \(2022\)](#): estimates a long-run neutral rate from surveys for long-horizon GDP growth and inflation expectations.

Financial term-structure models

- [Adrian-Crumb-Moench \(2013\)](#): primarily a model of the term premium, it delivers a nominal 'risk-neutral' short-term interest rate that's often used as a proxy to r^* . We follow the common approach and use the 10-year term premium to calculate the corresponding 10-year risk-neutral rate. A drawback is that this is the average of expected short-term rates over the next 10 years, rather than a neutral rate. To get to real rates, we subtract the 10-year break-even inflation rate.
- [Christensen-Mouabbi \(2024\)](#): new yield curve model estimated directly on the prices of bonds that are indexed to euro-area inflation. In contrast to other term-structure models, it estimates a real neutral rate, not a nominal risk-neutral rate.
- [Hördahl-Tristani \(2014\)](#): structural vector-autoregression on a broad range of variables, based on a joint model of macroeconomic and term structure dynamics.

Surveys

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- **ECB survey of monetary analysts**: new survey started in June 2021. We use the median long-term forecast of the deposit facility rate, and subtract the median long-term inflation forecast.
- **Survey of professional forecasters**: we use the mean longer-term interest rate, but since this series only starts in 2025, we extend it back with the mean 3-year ahead expected short-term rate. We subtract the corresponding mean longer-term inflation forecast, or the mean 3-year ahead inflation forecast.

Acknowledgments: We're grateful to Sarah Mouabbi for providing updated estimates of the Christensen-Mouabbi (2024) model.

Christensen, J.H.E., and Mouabbi, S. (2024) 'The Natural Rate of Interest in the Euro Area: Evidence from Inflation-Indexed Bonds', Federal Reserve Bank of San Francisco, Working Paper Series, 2024(08).